

MYSQL PROJECT

TOPIC: AMAZON PRODUCT DATA ANALYSIS USING SQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of an Amazon product dataset using MySQL. The dataset contains product-level information including product names, categories, actual and discounted prices, discount percentages, customer ratings, rating counts, and customer reviews.

The objective of this analysis is to extract meaningful insights related to product pricing, discounts, customer ratings, product popularity, and category performance using SQL techniques such as filtering, aggregation, grouping, sorting, calculations, HAVING clauses, and subqueries.

TOOL: MySQL Workbench

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Queries

1. Find the 10 products with the highest discounted price.

```
select product_id, product_name ,category,discouted_price
from amazon_clean order by discounted_price desc limit 10;
```

product_id	product_name	category	discounted_price
B07KRCW6LZ	TP-Link Nano AC600 USB Wi-Fi Adapter(Archer ...	Computers&Accessories NetworkingDevices Ne...	999
B0993BB11X	Ambrane 10000mAh Slim Power Bank, 20W Fast...	Electronics Mobiles&Accessories MobileAccessor...	999
B082T6GXS5	AmazonBasics New Release Nylon USB-A to Ligh...	Computers&Accessories Accessories&Peripheral...	999
B00O2R38C4	Luminous Vento Deluxe 150 mm Exhaust Fan fo...	Home&Kitchen Heating,Cooling&AirQuality Fans...	999
B084MZXJN6	Belkin Apple Certified Lightning to USB Charge a...	Computers&Accessories Accessories&Peripheral...	999
B07989VV5K	Usha Steam Pro SI 3713, 1300 W Steam Iron, P...	Home&Kitchen Kitchen&HomeAppliances Vacuu...	999
B094G9L9LT	KENT 16088 Vogue Electric Kettle 1.8 Litre 1500...	Home&Kitchen Kitchen&HomeAppliances SmallKi...	999
B0123P3PWE	Rico IRPRO 1500 Watt Japanese Technology El...	Home&Kitchen Heating,Cooling&AirQuality Wat...	999
B08TGG316Z	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ult...	Electronics HomeTheater,TV&Video Accessories...	999
B07H5PBN54	INDIASTM Electro-Instant Water Geyser A.B.S....	Home&Kitchen Heating,Cooling&AirQuality Wat...	999

Insight: This analysis identifies the 10 products with the highest discounted prices. It helps determine which products have relatively high selling prices even after discounts are applied.

2.What is the average customer rating for each product category

```
select category,avg(rating) as average_rating from
amazon_clean group by category limit 15;
```

AMAZON PRODUCT DATA ANALYSIS USING SQL

Result Grid	Filter Rows:	Export:	Wrap
category	average_rating		
Computers&Accessories Accessories&Peripheral...	4.147598253275108		
Computers&Accessories NetworkingDevices Ne...	3.9692307692307693		
Electronics HomeTheater,TV&Video Accessories...	4.254166666666668		
Electronics HomeTheater,TV&Video Accessories...	3.825		
Electronics HomeTheater,TV&Video Accessories...	4.300000000000001		
Electronics HomeAudio Accessories SpeakerAcc...	4		
Electronics HomeTheater,TV&Video Accessories...	4.2		
Electronics HomeAudio Accessories Adapters	4.4		
Electronics HomeTheater,TV&Video Accessories...	4		
Computers&Accessories Accessories&Peripheral...	4.4		
Electronics HomeTheater,TV&Video Accessories...	4.4		
Electronics HomeTheater,TV&Video SatelliteEqu...	4.2		
Electronics Accessories MemoryCards MicroSD	4.311111111111111		
Electronics Headphones,Earbuds&Accessories ...	3.9387096774193546		
Electronics Mobiles&Accessories MobileAccessor...	4.180000000000001		

Insight: This analysis compares the average customer ratings across different product categories. It helps identify categories with relatively higher or lower levels of customer satisfaction.

3.Which products have a rating of 4.5 or higher and have received more than 1,000 ratings?

```
select product_id,product_name from amazon_clean where  
rating>=4.5 and cast(replace(rating_count,',','') as unsigned)>1000  
limit 15;
```

AMAZON PRODUCT DATA ANALYSIS USING SQL

Result Grid	Filter Rows:	Export:
product_id	product_name	
B00NH11PEY	AmazonBasics USB 2.0 - A-Male to A-Female Ex...	
B00NH11KIK	AmazonBasics USB 2.0 Cable - A-Male to B-Male...	
B08TGG316Z	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ult...	
B082T6GXS5	AmazonBasics New Release Nylon USB-A to Ligh...	
B00NH13Q8W	AmazonBasics USB 2.0 Extension Cable for Pers...	
B07RY2X9MP	AmazonBasics 10.2 Gbps High-Speed 4K HDMI ...	
B0116MIKKC	Goldmedal Curve Plus 202042 Plastic Spice 3-Pin...	
B095RTJH1M	Spigen EZ Fit Tempered Glass Screen Protector ...	
B0823LW7NV	Spigen EZ Fit Tempered Glass Screen Protector ...	
B08244R4KB	Spigen EZ Fit Tempered Glass Screen Protector ...	
B07Q4QV1DL	ELV Aluminum Adjustable Mobile Phone Foldable...	
B088ZFJY82	Elv Aluminium Adjustable Mobile Phone Foldable ...	
B01HJIOFS2	Dell MS116 1000Dpi USB Wired Optical Mouse, L...	
B014SZO90Y	Duracell Ultra Alkaline AA Battery, 8 Pcs	
B0765B3TH7	Gizga Essentials Hard Drive Case Shell, 6.35cm/...	

Insight: This analysis identifies products that have both a rating of 4.5 or higher and more than 1,000 ratings. These products can be considered highly rated and sufficiently reviewed by customers.

4.Which 10 products offer the highest discount percentage?

```
select product_id,product_name,discount_percentage from  
amazon_clean order by cast(replace(discount_percentage,'%','') as  
unsigned) desc limit 10;
```

AMAZON PRODUCT DATA ANALYSIS USING SQL

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
product_id	product_name	discount_percentage	
B097C564GC	rts [2 Pack] Mini USB C Type C Adapter Plug, Ty...	94%	
B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync...	90%	
B085HY1DGR	Sounce Spiral Charger Cable Protector Data Ca...	90%	
B09SJ1FTYV	Sounce Protective Case Cover Compatible Boat ...	90%	
B08W56G1K9	LAPSTER Spiral Charger Spiral Charger Cable Pr...	90%	
B08X77LM8C	Silicone Rubber Earbuds Tips, Eartips, Earpads, ...	90%	
B0861DSF17	beatXP Kitchen Scale Multipurpose Portable Elec...	90%	
B085HY1DGR	Sounce Spiral Charger Cable Protector Data Ca...	90%	
B08BW521YC	LAPSTER 12pcs Spiral Cable Protectors for Char...	90%	
B084T6MR8N	pTron Solero M241 2.4A Micro USB Data & Char...	89%	

Insight: This analysis identifies the 10 products with the highest discount percentages. It helps determine which products provide the most significant percentage-based discounts to customers.

5. What is the average actual price and average discounted price for each category?

```
select category, avg(cast(replace(actual_price, ',', '')) as unsigned)) as  
average_actual_price, avg(discounted_price) as  
average_discounted_price from amazon_clean group by category  
limit 15;
```

AMAZON PRODUCT DATA ANALYSIS USING SQL

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows
category	average_actual_price	average_discounted_price		
Computers&Accessories Accessories&Peripheral...	887.3624	341.1921		
Computers&Accessories NetworkingDevices Ne...	1015.1538	492.5385		
Electronics HomeTheater,TV&Video Accessories...	1175.2500	406.2917		
Electronics HomeTheater,TV&Video Accessories...	951.2955	332.2273		
Electronics HomeTheater,TV&Video Accessories...	979.0000	464.0000		
Electronics HomeAudio Accessories SpeakerAcc...	1299.0000	349.0000		
Electronics HomeTheater,TV&Video Accessories...	1299.0000	451.0000		
Electronics HomeAudio Accessories Adapters	600.0000	209.0000		
Electronics HomeTheater,TV&Video Accessories...	812.2500	446.7500		
Computers&Accessories Accessories&Peripheral...	1100.0000	499.0000		
Electronics HomeTheater,TV&Video Accessories...	795.0000	399.0000		
Electronics HomeTheater,TV&Video SatelliteEqu...	2299.0000	917.0000		
Electronics Accessories MemoryCards MicroSD	1533.2222	599.0000		
Electronics Headphones,Earbuds&Accessories ...	1491.1613	542.6129		
Electronics Mobiles&Accessories MobileAccessor...	1199.0000	495.8000		

Insight: This analysis compares the average actual price and average discounted price for each category. It helps understand pricing patterns and how discounts affect product prices across different categories.

6.Which products have a price difference of more than ₹1,000 between their actual price and discounted price?

```
select product_id,product_name ,actual_price,discouted_price from  
amazon_clean where cast(replace(actual_price,',','') as unsigned) -  
discouted_price >1000 limit 15;
```

AMAZON PRODUCT DATA ANALYSIS USING SQL

product_id	product_name	actual_price	discounted_price
B096MSW6CT	Source Fast Phone Charging Cable & Data Sync...	1,899	199
B09W5XR9RT	Duracell USB C To Lightning Apple Certified (Mfi)...	1,999	970
B07XLCFSSN	Amazonbasics Nylon Braided Usb-C To Lightning...	1,900	899
B09C6HWG18	Duracell Type C To Type C 5A (100W) Braided S...	1,999	970
B082T6V3DT	AmazonBasics New Release Nylon USB-A to Ligh...	2,100	799
B0B4DT8MKT	Wecool Unbreakable 3 in 1 Charging Cable with ...	1,499	348
B014I8SX4Y	Amazon Basics High-Speed HDMI Cable, 6 Feet ...	1,400	309
B07JH1C41D	Wayona Nylon Braided (2 Pack) Lightning Fast ...	1,999	649
B09BW334ML	Dealfreez Case Compatible with Fire TV Stick 3r...	1,499	349
B09TT6BFDX	Cotbolt Silicone Protective Case Cover for LG a...	1,999	399
B08TGG316Z	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ult...	2,399	999
B082T6GXS5	AmazonBasics New Release Nylon USB-A to Ligh...	2,100	999
B09JKNF147	Electvision Remote Control Compatible with Kod...	1,999	339
B0BP7XLX48	Syncwire LTG to USB Cable for Fast Charging C...	1,999	399
B08XMSKKMM	7SEVEN Bluetooth Voice Command Remote for X...	1,999	799

Insight: This query identifies products where the difference between the original price and discounted price is greater than ₹1,000. It helps highlight products that provide a substantial amount of monetary savings to customers.

7.Which 10 products have received the highest number of ratings?

```
select product_id,product_name,rating_count from amazon_clean
order by cast(replace(rating_count,',','') as unsigned) desc limit 10;
```

product_id	product_name	rating_count
B014I8SX4Y	Amazon Basics High-Speed HDMI Cable, 6 Feet ...	4,26,973
B014I8SSD0	Amazon Basics High-Speed HDMI Cable, 6 Feet ...	4,26,973
B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Bla...	4,26,973
B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Bla...	4,26,972
B07GPXXNNG	boAt Bassheads 100 in Ear Wired Earphones wit...	3,63,713
B07GQD4K6L	boAt Bassheads 100 in Ear Wired Earphones wit...	3,63,713
B071Z8M4KX	boAt BassHeads 100 in-Ear Wired Headphones ...	3,63,711
B01MF8MB65	boAt Bassheads 225 in Ear Wired Earphones wit...	2,73,189
B01LWYDEQ7	Pigeon Polypropylene Mini Handy and Compact ...	2,70,563
B005FYNT3G	SanDisk Cruzer Blade 32GB USB Flash Drive	2,53,105

Insight: This analysis identifies the 10 products with the highest number of customer ratings. A high rating count indicates that these products have received considerable customer attention and engagement.

8. Which categories have an average rating greater than 4.0?

```
select category, avg(rating) as average_rating from amazon_clean
group by category having avg(rating) > 4.0 limit 15;
```

Result Grid			Filter Rows:	Export:	Wrap
	category	average_rating			
▶	Computers&Accessories Accessories&Peripheral...	4.147598253275108			
	Electronics HomeTheater,TV&Video Accessories...	4.254166666666668			
	Electronics HomeTheater,TV&Video Accessories...	4.300000000000001			
	Electronics HomeTheater,TV&Video Accessories...	4.2			
	Electronics HomeAudio Accessories Adapters	4.4			
	Computers&Accessories Accessories&Peripheral...	4.4			
	Electronics HomeTheater,TV&Video Accessories...	4.4			
	Electronics HomeTheater,TV&Video SatelliteEqu...	4.2			
	Electronics Accessories MemoryCards MicroSD	4.311111111111111			
	Electronics Mobiles&Accessories MobileAccessor...	4.180000000000001			
	Electronics Mobiles&Accessories MobileAccessor...	4.25			
	Electronics Mobiles&Accessories MobileAccessor...	4.033333333333333			
	Electronics Mobiles&Accessories MobileAccessor...	4.107142857142857			
	Electronics Mobiles&Accessories MobileAccessor...	4.16			
	Computers&Accessories Accessories&Peripheral...	4.033333333333333			

Insight: This analysis identifies product categories with an average customer rating above 4.0. These categories show relatively positive customer feedback based on their average ratings.

9.What is the average customer rating of products offering a discount of 50% or more?

```
select avg(rating) as average_rating from amazon_clean where  
cast(replace(discount_percentage,'%','')as unsigned)  
>=50;
```

Result Grid		Filter Rows:
	average_rating	
▶	4.064088397790051	

Insight: This analysis calculates the average customer rating of products offering discounts of 50% or more. It helps examine the level of customer satisfaction among products with relatively high discounts.

10.What are the 10 most expensive products based on their actual price?

```
select product_id,product_name,actual_price from amazon_clean  
order by cast(replace(actual_price,',','') as unsigned) desc limit 10;
```

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product_id	product_name	actual_price
B097C564GC	rtts [2 Pack] Mini USB C Type C Adapter Plug, Ty...	4,999
B0718ZN31Q	RtsTM High Speed 3D Full HD 1080p Support (1...	4,999
B08MZQBFLN	Callas Multipurpose Foldable Laptop Table with ...	4,999
B07LG59NPV	Boult Audio Probass Curve Bluetooth Wireless in...	4,499
B0B1F6GQPS	Boult Audio FXCharge with ENC, 32H Playtime, ...	4,499
B0B2931FCV	ZEBRONICS Zeb-Sound Bomb N1 True Wireless i...	4,199
B09Z28BQZT	Amazon Basics Multipurpose Foldable Laptop Ta...	3,999
B0B54Y2SNX	iPhone Original 20W C Type Fast PD Charger C...	3,990
B07N42JB4S	SYVO WT 3130 Aluminum Tripod (133CM), Univ...	3,990
B0B82YGCF6	Tokdis MX-1 Pro Bluetooth Calling Smartwatch - ...	3,499

Insight: This analysis identifies the 10 products with the highest actual prices before discounts. It helps determine which products belong to the higher-priced segment of the dataset.

11.Which product category has the highest average discount percentage?

```
select category,avg(cast(replace(discount_percentage,'%','') as unsigned)) as average_discount_percentage from amazon_clean group by category order by average_discount_percentage desc limit 1;
```

category	average_discount_percentage
Computers&Accessories Accessories&Peripheral...	90.0000

Insight: This analysis compares the average discount percentage across product categories and identifies the category with the highest average discount. It helps determine which category offers the greatest discounts on average.

12. Which products have both a rating of at least 4.0 and more than 5,000 ratings?

```
select product_id, product_name, rating, rating_count from
amazon_clean where rating >= 4.0 and CAST(replace(rating_count, ',',
'') as unsigned) > 5000 limit 15;
```

Result Grid				
		Filter Rows:		Export:
		Wrap Cell Content:		
	product_id	product_name	rating	rating_count
▶	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Ch...	4.2	24,269
	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging ...	4	43,994
	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB ...	4.2	94,363
	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	4.2	16,905
	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro US...	4.1	15,188
	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	4.3	30,411
	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), ...	4.2	1,79,691
	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging ...	4	43,994
	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3...	4.3	13,391
	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braid...	4.2	94,363
	B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Bla...	4.4	4,26,973
	B083342NKJ	MI Braided USB Type-C Cable for Charging Ada...	4.4	18,757
	B082LSVT4B	Ambrane Unbreakable 60W / 3A Fast Charging ...	4	43,994
	B08WRBG3XW	boAt Type C A325 Tangle-free, Sturdy Type C ...	4.1	13,045
	B09NHVCHS9	Flix Micro Usb Cable For Smartphone (Black)	4	9,378

Insight: This analysis identifies products that have a rating of at least 4.0 and more than 5,000 ratings. These products can be considered both well-rated and highly reviewed by customers.

13. How much money can customers save on each product by comparing the actual price with the discounted price?

```
select product_id, product_name, CAST(replace(actual_price, ',', '')) as
unsigned) - discounted_price as money_saved from amazon_clean
limit 15;
```

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Content	
	product_id	product_name	money_saved
▶	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Ch...	700
	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging ...	150
	B096MSW6CT	Source Fast Phone Charging Cable & Data Sync...	1700
	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB ...	370
	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	245
	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast C...	851
	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro US...	322
	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	70
	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), ...	500
	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging ...	100
	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3...	185
	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braid...	500
	B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Bla...	481
	B085DTN6R2	Portronics Konnect CL 20W POR-1067 Type-C t...	549
	B09KLVNZ3B	Portronics Konnect L 1.2M POR-1401 Fast Char...	240

Insight: This analysis calculates the amount of money saved on each product by comparing its actual price with its discounted price. It helps identify the monetary benefit customers receive from product discounts.

14. Which products have a rating higher than the overall average rating of all products?

```
select product_id,product_name,rating from amazon_clean where
rating>(select avg(rating) from amazon_clean) limit 15;
```

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cells	
	product_id	product_name	rating
▶	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Ch...	4.2
	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB ...	4.2
	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	4.2
	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro US...	4.1
	B08DDRGTJ	MI Usb Type-C Cable Smartphone (Black)	4.3
	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), ...	4.2
	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3...	4.3
	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braid...	4.2
	B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Bla...	4.4
	B085DTN6R2	Portronics Konnect CL 20W POR-1067 Type-C t...	4.2
	B09KLMZ3B	Portronics Konnect L 1.2M POR-1401 Fast Char...	4.1
	B083342NKJ	MI Braided USB Type-C Cable for Charging Ada...	4.4
	B08WRBG3XW	boAt Type C A325 Tangle-free, Sturdy Type C ...	4.1
	B09C6HXFC1	Duracell USB Lightning Apple Certified (Mfi) Brai...	4.5
	B01M4GGIVU	Tizum High Speed HDMI Cable with Ethernet S...	4.2

Insight: This analysis compares individual product ratings with the overall average rating of all products. It identifies products that perform above the overall rating level of the dataset.

15.Which product categories have both an average rating above 4.0 and more than 10 products?

```
select category,avg(rating) as average_rating,count(*) as  
product_count    from amazon_clean  
group by category having avg(rating)>4.0 and count(*)>10;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category	average_rating	product_count	
Computers&Accessories Accessories&Peripheral...	4.147598253275108	229	
Electronics HomeTheater,TV&Video Accessories...	4.254166666666668	24	
Electronics Mobiles&Accessories MobileAccessor...	4.107142857142857	14	
Computers&Accessories Accessories&Peripheral...	4.25	20	
Computers&Accessories Accessories&Peripheral...	4.046153846153846	13	
Home&Kitchen Kitchen&HomeAppliances Vacuu...	4.02	20	
Home&Kitchen Kitchen&HomeAppliances Vacuu...	4.121052631578947	19	

Insight: This analysis identifies product categories that have an average rating above 4.0 and contain more than 10 products. It helps identify categories that combine positive customer feedback with a sufficient number of products.

CONCLUSION

The **Amazon Product Data Analysis** project successfully demonstrated the use of MySQL for analyzing e-commerce product data and generating meaningful insights. Through the application of SQL concepts such as filtering, aggregation, GROUP BY, HAVING, sorting, calculations, and subqueries, the project transformed raw Amazon product records into useful information.

The analysis provided valuable insights into **product pricing, discount patterns, customer ratings, product popularity, and category performance**. Highly rated and frequently reviewed products were identified, while products offering significant percentage and monetary discounts were also analyzed. The comparison between actual and discounted prices helped determine the potential savings available to customers.

Additionally, the analysis highlighted product categories based on their **average ratings, average prices, and average discount percentages**. This helped in understanding differences in customer satisfaction, pricing strategies, and discount levels across categories.

Overall, this project strengthened practical SQL skills and demonstrated how SQL can be applied to **real-world e-commerce data** to identify patterns, compare products, and generate meaningful insights. The project also showed how data analysis can support better understanding of **customer preferences, product pricing, discounts, and product performance** for data-driven decision-making.

